

Curriculum Vitae

Yewen (Evan) Pu | 濮燚文

yewen.pu@ntu.edu.sg [work] // yewenpu1989@gmail.com [personal]

<https://evanthebouncy.github.io/>

Employment

Nanyang Technological University

2025 -

Assistant Professor

- code generation for instruction following
- multi-turn task alignment for agents

Autodesk Research

Principal Research Scientist

- code generation for computer-aided designs (CAD)
- industry scale multi-modal human data curation for instruction tuning for CAD

Massachusetts Institute of Technology

2019 - 2020

Postdoctoral Researcher

- program synthesis and computational cognitive science

Massachusetts Institute of Technology

2012 - 2019

Graduate Student Researcher

- program synthesis and active learning

Learable.ai

2018 - 2019

Machine Learning Advisor

- ML + building data annotation pipeline for ML

Microsoft Research

2018 summer

Research Intern

Education

Massachusetts Institute of Technology

2019

PhD of Computer Science

Advisors: Armando Solar-Lezama, Leslie Kaelbling

University of California, Berkeley

2011

B.A. of Computer Science

B.A. of Mathematics (with high honor)

Selected Publications

Cogsci 2026	Cassidy Langenfeld, Claas Beger, Gloria Geng, Wasu Top Piriyaakulkij, Keya Hu, <u>Yewen Pu</u> , Kevin Ellis Bongards at the Boundary of Perception and Reasoning: Programs or Language?
CVPR 2026	Tao Wu, Chuha Zhou, Guangyu Zhao, Haozhi Cao, <u>Yewen Pu</u> , Jianfei Yang When Robots Should Say "I Don't Know": Benchmarking Abstention in Embodied Question Answering
EMNLP 2025	William P. McCarthy, Saujas Vaduguru, Karl D. D. Willis, Justin Matejka, Judith E. Fan, Daniel Fried, <u>Yewen Pu</u> mrCAD: Multimodal Refinement of Computer-aided Designs
ICLR 2025	Wen-Ding Li, Keya Hu, Carter Larsen, Yuqing Wu, Simon Alford, Caleb Woo, Spencer M Dunn, Hao Tang, Michelangelo Naim, Dat Nguyen, Wei-Long Zheng, Zenna Tavares, <u>Yewen Pu</u> , Kevin Ellis Combining induction and transduction for abstract reasoning
ECCV 2024	Sifan Wu, Amir Hosein Khasahmadi, Mor Katz, Pradeep Kumar Jayaraman, <u>Yewen Pu</u> , Karl Willis, Bang Liu CadVLM: Bridging Language and Vision in the Generation of Parametric CAD Sketches
C&C 2024	William McCarthy, Justin Matejka, Karl Willis, Judith Fan, <u>Yewen Pu</u> Communicating Design Intent Using Drawing and Text
ICML 2024	<u>Yewen Pu</u> , Saujas Vaduguru, Priyan Vaithilingam, Elena Glassman, Daniel Fried Amortizing Pragmatic Program Synthesis with Rankings
ICLR 2024	Ruocheng Wang, Eric Zelikman, Gabriel Poesia, <u>Yewen Pu</u> , Nick Haber, Noah D Goodman Hypothesis search: Inductive reasoning with language models
ICLR 2024	Saujas Vaduguru, Daniel Fried, <u>Yewen Pu</u> Generating Pragmatic Examples to Train Neural Program Synthesizers
NeurIPS 2023	Di Huang, Ziyuan Nan, Xing Hu, Pengwei Jin, Shaohui Peng, Yuanbo Wen, Rui

	Zhang, Zidong Du, Qi Guo, <u>Yewen Pu</u> , Yunji Chen ANPL: Towards Natural Programming with Interactive Decomposition
NeurIPS 2023	Zhao Huang, Feng Chen, <u>Yewen Pu</u> , Chunru Lin, Hao Su, Chuang Gan DiffVL: scaling up soft body manipulation using vision-language driven differentiable physics
NeurIPS 2022 (Oral)	Sam Acquaviva*, <u>Yewen Pu</u> *, Marta Kryven, Theodoros Sechopoulos, Catherine Wong, Gabrielle Ecanow, Maxwell Nye, Michael Tessler, Josh Tenenbaum Communicating natural programs to humans and machines
SIGGRAPH 2021	Karl DD Willis, <u>Yewen Pu</u> , Jieliang Luo, Hang Chu, Tao Du, Joseph G Lambourne, Armando Solar-Lezama, Wojciech Matusik Fusion 360 gallery: A dataset and environment for programmatic cad construction from human design sequences
CVPR 2021	Karl DD Willis, Pradeep Kumar Jayaraman, Joseph G Lambourne, Hang Chu, <u>Yewen Pu</u> Engineering sketch generation for computer-aided design
NeurIPS 2021	Yichen Yang, Jeevana Priya Inala, Osbert Bastani, <u>Yewen Pu</u> , Armando Solar-Lezama, Martin Rinard Program synthesis guided reinforcement learning for partially observed environments
ICLR 2021	Maxwell Nye, <u>Yewen Pu</u> , Matthew Bowers, Jacob Andreas, Joshua B Tenenbaum, Armando Solar-Lezama Representing partial programs with blended abstract semantics
NeurIPS 2020	Jeevana Priya Inala, Yichen Yang, James Paulos, <u>Yewen Pu</u> , Osbert Bastani, Vijay Kumar, Martin Rinard, Armando Solar-Lezama Neurosymbolic Transformers for Multi-Agent Communication
NeurIPS 2020	<u>Yewen Pu</u> , Kevin Ellis, Marta Kryven, Josh Tenenbaum, Armando Solar-Lezama Program synthesis with pragmatic communication

NeurIPS 2019	Kevin Ellis*, Maxwell Nye*, <u>Yewen Pu</u> *, Felix Sosa*, Josh Tenenbaum, Armando Solar-Lezama Write, execute, assess: Program synthesis with a repl
NeurIPS 2019	Charith Mendis, Cambridge Yang, <u>Yewen Pu</u> , Dr Saman Amarasinghe, Michael Carbin Compiler auto-vectorization with imitation learning
NeurIPS 2018	Osbert Bastani, <u>Yewen Pu</u> , Armando Solar-Lezama Verifiable reinforcement learning via policy extraction
SIGGRAPH Asia 2018	Tao Du, Jeevana Priya Inala, <u>Yewen Pu</u> , Andrew Spielberg, Adriana Schulz, Daniela Rus, Armando Solar-Lezama, Wojciech Matusik Inversecsg: Automatic conversion of 3d models to csg trees
ICML 2018 (Oral)	<u>Yewen Pu</u> , Zachery Miranda, Armando Solar-Lezama, Leslie Kaelbling Selecting representative examples for program synthesis
UAI 2017	<u>Yewen Pu</u> , Leslie P Kaelbling, Armando Solar-Lezama Learning to acquire information
OOPSLA 2016 (workshop)	<u>Yewen Pu</u> , Karthik Narasimhan, Armando Solar-Lezama, Regina Barzilay sk_p: a neural program corrector for MOOCs
POPL 2013	Ali Koksai, <u>Yewen Pu</u> , Saurabh Srivastava, Rastislav Bodik, Jasmin Fisher, Nir Piterman Synthesis of biological models from mutation experiments
OOPSLA 2011	<u>Yewen Pu</u> , Rastislav Bodik, Saurabh Srivastava Synthesis of first-order dynamic programming algorithms

Preprints

arxiv 2024	Xianghao Xu, Aditya Ganeshan, Karl DD Willis, <u>Yewen Pu</u> , Daniel Ritchie Creating Language-driven Spatial Variations of Icon Images
------------	--

arxiv 2023	Leonardo Hernandez Cano, <u>Yewen Pu</u> , Robert Hawkins, Josh Tenenbaum, Armando Solar-Lezama
	Building Community Driven Libraries of Natural Programs
arxiv 2023	Priyan Vaithilingam, <u>Yewen Pu</u> , Elena L Glassman
	The Usability of Pragmatic Communication in Regular Expression Synthesis

Invited Talks

Predicting Performance of Symbolic and Prompt Programs with Examples @ AAAI Trustworthy Agentic AI workshop Keynote	2026 Jan
mrCAD: Multimodal Refinement of Computer-aided Designs @ Physics-Driven AI Engineering Design Workshop	2026 Jan
mrCAD: Multimodal Refinement of Computer-aided Designs @ Dagstuhl Seminar on Inductive Programming	2025 Dec
Human-AI collaborative iteration @ Tsinghua	2025 Oct
Curating human-human instruction dataset @ MIT CAP group	2025 May
Curating human-human instruction dataset @ Berkeley NLP Lunch	2025 April
Curating human-human instruction dataset @ Basis.ai	2025 April
Curating human-human instruction dataset @ CMU Language Technologies Institute Colloquium	2024 Nov
Code Generation Inspired by Human Communication @ SUTD	2024 Oct
Code Generation Inspired by Human Communication @ NUS	2024 Oct
Code Generation Inspired by Human Communication @ NTU	2024 Oct
Code Generation Inspired by Human Communication @ Stanford CogToolsLab	2024 Aug

Pragmatic Program Synthesis @ Boston Computation Club	2024 July
Code Generation Inspired by Human Communication @ Berkeley NLP Lunch	2024 July
Pragmatic Program Synthesis @ UCSD Design Lab Research Invited Guest Talk	2024 May
Pragmatic Program Synthesis @ Vienna University of Technology	2024 May
Pragmatic Program Synthesis @ Cocosci Lunch / MIT	2024 May
Pragmatic Program Synthesis @ Neuro-Symbolic Seminar / MIT	2024 March
Pragmatic Program Synthesis @ UCSD PL Seminar	2024 Feb
Natural Programming in Communicating Designs @ 2023 Frontiers in Design Representation Summer School	2023 June
Programming as Collaborative Search @ Stanford cocolab	2023 April
Communicating Natural Programs to Humans and Machines @ carper.ai	2022 Nov
LARC : Language annotated Abstraction and Reasoning Corpus @ MIT cocosci	2021 April
Program Synthesis with Pragmatic Communication @ MIT Cpl-lab	2021 Jan
Program Synthesis with Pragmatic Communication @ Ras Bodik Group @ University of Washington	2020 Nov
Program Synthesis with Pragmatic Communication @ PL Seminar @ UCSD	2020 Oct
Program Synthesis with Pragmatic Communication @ Stanford cocolab	2020 Aug
Program Synthesis with Pragmatic Communication @ Variation Lab @ Harvard SEAS	2020 June
Program Synthesis as Communication @ Autodesk	2020 March

Program Synthesis as Communication @ Deepmind	2019 Dec
Program Synthesis : From Communication to Computation @ Machine Programming Group @ Intel Labs	2019 Nov
Write, Execute, Assess: Program Synthesis with a REPL @ Google Brain	2019 Oct
Write, Execute, Assess: Program Synthesis with a REPL @ Facebook	2019 Oct

Grants

PI, Nanyang Assistant Professorship, "Learning to Follow Instructions through Human Interactions", 1.5M SGD | NTU internal

PI, Expert-in-the-Loop Agentic AI for Knowledge Discovery and Evolution in Manufacturing
228,000 SGD | ARTC (A*STAR Research Entities)

Advising

Shuzhi Liu (PhD)	2025 -
Tao Wu (PhD)	2026 -
Chengqi Zheng (PhD)	2026 -
DucAnh Do (Master)	2026 -
Abhipray Chavan (Master)	2026 -
Yan Li (PostDoc)	2026 -

External Advising

(MIT) Leonardo Hernández Cano, with Armando Solar-Lezama	2020 -
--	--------

(CMU) Saujas Vaduguru, with Daniel Fried	2021 - 2025
(CAS) Di Huang	2022 - 2023
(Berkeley) Rodolfo (Rudy) Corona, with Dan Klein	2024 - 2024
(Cornell) WenDing Li, with Kevin Ellis	2024 - 2024
(Cornell) Keya Hu, with Kevin Ellis	2024 - 2025
(Stanford) Rio Aguina-kang, with Judith Fan	2024 - 2024
(Brown) Xianghao Xu, with Daniel Ritchie	2023 - 2024
(UCSD) William McCarthy, with Judith Fan	2023 - 2024
(MIT) Sam Acquaviva, with Josh Tenenbaum	2020 - 2024
(Stanford) Ruocheng Wang, with Noah Goodman	2023 - 2023
(UCSD) Zhiao Huang, with Chuang Gan	2023 - 2023
(Harvard) Priyan Vaithilingam, with Elena Glassman	2022 - 2023

Services

Thesis Committee

Samuel Acquaviva, 2024 (Master)	Massachusetts Institute of Technology
Fernando Abigail Galicia Mendoza, 2023 (PhD)	Universidad Nacional Autónoma de México

Reviewer

NeurIPS	2018 -
ICML	2018 -
ICLR	2020 -
CogSci	2022 -
AAAI	2023 -

UIST	2023 -
ACL / EMNLP	2023 -
UAI	2018